AP PHYSICS 1 IA-1 — COMPLETE EXTRACTED DATA (Public Archive)

Everything extracted from the three sources: **Newton School Portal**, **Slack**, **WhatsApp hardcopy photos (146)**.

Compiled 26 Oct 2026. Exam: Applied Physics 1 IA-1, Tue 27 Oct 2026, 9:40 AM.

SOURCE 1 — PORTAL (my.newtonschool.co, course tjj12ybxo9tt)

Course: NSTB'26-CS, subject **AP-Robo B** = Applied Physics 1. Parent course kxz81gy1l9wm.

My performance (Overview): Lectures 83% (10/12) · Assignments 0% (0/0) · Contests 50% (1/2) · Quizzes 0% (0/0). Total XP 5,795.

Lectures (13 total):

#	Date	Topic	Status
L1	31 Aug	Measurement Fundamentals	attended
L2	2 Sep	Scalars and Vectors, Vector Arithmetic, Vector Products, Vector Resolution	attended
L3	7 Sep	Kinematics, Distance and Displacement, Speed and Velocity, Acceleration	attended
L4	9 Sep	Relative Motion, Solving 1-D Motion, Projectile Motion, Breaking Down Multi-D Motion	attended, notes
L5	16 Sep	NLMs & FBDs, Forces & Types, Newton's Laws, Frames of Reference, Pseudo Forces	notes, VIDEO UNWATCHED (0/30)
L6	21 Sep	Friction, Normal & Coefficient of Friction, Friction & Types, Multi-D FBDs	attended, notes
L7	23 Sep	Circular Motion, Angular Coordinate System, Radial/Tangential Vectors	VIDEO UNWATCHED (0/30)
L8	28 Sep	Work and Energy	attended
L9	30 Sep	Circular Motion / CM	attended, notes
L10	5 Oct	Circular Motion / CM	attended, notes
L11	7 Oct	(Circular Motion)	VIDEO UNWATCHED (0/30)
L12	12 Oct	(Circular Motion)	attended
L13	14 Oct	(Circular Motion)	attended

Note: **IA-1 syllabus stops at L10.** L11–13 are outside the exam.

Contests:

- Assignment Contest 1 — solved (1/2 counted).
- **Applied Physics 1 - Contest 2 (Subjective)** — released 16 Oct, deadline 16 Oct 11:20 AM, topic **Centre of Mass**, difficulty easy, **0/40 XP earned**, solved by 34 students (15%). STILL OPEN BACKLOG.

Assignments: none found (0/0). **Quiz:** 0/0 (1 historical quiz contest, completed).

Contest-2 question text (playground/subjective/9gkqeb7ttnk6): an ice cube (mass m, side L) melts in a tray (mass M); the water spreads into a thin layer. Does the centre of mass of the system move? Explain without

numerical calculation: how the redistribution of mass affects it, and how the relative masses m and M influence the magnitude of the change.

SOURCE 2 — SLACK (#academic-announcement-26)

Official AP 1 IA-1 syllabus (nikhil.menon, 14 Oct 2026, 5:14 PM) — VERBATIM, "From Lecture 1 – Lecture 10":

Sizes in the Universe · Introduction to Measurement · Units and Dimensions · Order of Magnitude · Significant Digits · Scientific Notation · Error and Propagation · Scalars and Vectors · Vector Arithmetic · Vector Products · Vector Resolution · Differentiation · Integration · Distance and Displacement · Speed and Velocity · Acceleration and Jerk · Graphical Understanding of Motion · Motion in Straight Line · Equations of Motion · Relative Motion · Solving 1-D Motion · Projectile Motion · Breaking Down Multi-D Motion · Forces & Types · Newton's Laws of Motion · Frames of Reference · Pseudo Forces & Inertia · Free Body Diagrams · Friction & Types · Normal & Coefficient of Friction · Multi-D FBDs with Friction · Multi-D NLM & Friction Problems · Angular Coordinate System · Radial and Tangential Vectors · CM: Angular Displacement & Speed · CM: Angular Velocity & Acceleration · CM: Centripetal & Centrifugal · Solving Circular Motion · Kinetic Energy · Potential Energy · Work Done · Spring Force · Conservative vs Non-Conservative Forces · Work and Energy Theorem · Centre of Mass · CoM: System of Objects · CoM: Distributed Systems · CoM: Multi-Dimensional Objects · Linear Momentum · Conservation of Linear Momentum · Conservation of LM: Multi-Dimensional · Conservation of LM: Multi-Stage · Collisions for Rigid Bodies · Elastic Collision · Inelastic Collision · Coefficient of Restitution · Multi-Dimensional Collisions.

Contest 2 syllabus (9 Oct 2026, 2:17 PM) — VERBATIM:

Forces & Types · Newton's Laws of Motion · Frames of Reference · Pseudo Forces & Inertia · Free Body Diagrams · Friction & Types · Normal & Coefficient of Friction · Multi-D FBDs with Friction · Multi-D NLM & Friction Problems · Angular Coordinate System · Radial and Tangential Vectors · CM: Angular Displacement & Speed · CM: Angular Velocity & Acceleration · CM: Centripetal & Centrifugal · Solving Circular Motion · Kinetic Energy · Potential Energy · Work Done · Spring Force · Conservative vs Non-Conservative Forces · Work and Energy Theorem · Centre of Mass · CoM: System of Objects · CoM: Distributed Systems · CoM: Multi-Dimensional Objects · Linear Momentum · Conservation of Linear Momemtum · Conservation of LM: Multi-Dimensional · Conservation of LM: Multi-Stage · Collisions for Rigid Bodies · Elastic Collision · Inelastic Collision · Coefficient of Restitution · Multi-Dimensional Collisions — "i.e From Worksheet 05 - 10".

Other AP facts from Slack:

- Contest 1: Fri 11 Sep (postponed to 1:30–3:30 PM), offline, classroom Athena. Syllabus = first four lectures (L1–L4).
- Contest 2: Fri 16 Oct, 9–11 AM, classroom Athena, subjective.
- Exam guidelines PDF + seating-plan PDFs shared as files.slack.com links.
- AP Learning Tracker: docs.google.com/spreadsheets/d/1ZOx2WQ11_7dymZ7tux25VkwUmWyXea1pGLauEx-WNIU (AP column).
- Also from Slack: IA-1 syllabi for PSP (programming) and other subjects exist in the same channel (out of scope here).

SOURCE 3 — WHATSAPP HARDCOPY PHOTOS (146 images, ap_media/m_001 … m_146)

Photo inventory (what each photo is)

- **m_004–m_006:** Maths IA-1 paper (BTNE113) — not AP.
- **Lecture notes:** m_007 (Measurement L01), m_015 (Vectors L02), m_032 (Motion L03), m_045–047 (projectile notes), m_001/m_003/m_048/m_052–053/m_055–056 (NLM/forces notes), m_076–078, m_084–085, m_092, m_100, m_105–113 (notes incl. two-stage momentum m_106, 2D momentum m_107), m_117–124, m_133–134, m_136–137.
- **Measurement worksheet:** m_143 (Q10–11), m_144 (Q6–9), m_145/m_146 (Q1–5); older copies m_012, m_014.
- **Calculus/1D worksheet:** m_025 (Q1–4), m_028 (Q11–12), m_029, m_042–044.
- **Projectile:** m_002, m_049. **Relative velocity:** m_050. **NLM:** m_051, m_054, m_057–059.
- **Friction worksheet:** m_140 (Q8–11), m_141 (Q5–7), m_142 (Q1–4).
- **Circular worksheet:** m_081–091 (single questions incl. Q1, Q6, Q11), m_127 (Q27–30), m_128 (Q22–26), m_129 (Q17–21), m_130 (Q7–11), m_131 (Q12–16), m_132 (Q1–6), m_135 (Q27–30).
- **Work & Energy worksheet:** m_116 (Q1–6), m_115 (Q7–12), m_114 (Q13–15).
- **Collisions worksheet:** m_075 (Q1–4), m_079 (Q13–15), m_072–074 (Q5–12 incl. "move together" stick cases).
- **Momentum worksheet:** m_101 (Q14–17), m_102 (Q10–13), m_103 (Q5–9).
- **CoM worksheet:** m_104 (Q1–4).
- **Contest-1 printed paper:** m_061 (Q28b airplane), m_062 (Q28a truck), m_063/m_064 (Q24, Q25), m_065 (Q21 errors, a(t), traffic), m_066 (Q18–20), m_067 (Q13–17), m_068 (Q3–7), m_069 (Q8–12), m_070–071 (covers).
- **Blank:** m_060, m_080.

ORIGINAL QUESTIONS — Worksheet: Measurement (g = 9.8)

1. Importance of measurement in engineering (explain, 3M).
2. Fundamental vs derived quantities; SI units of any 5 fundamental.
3. Write dimensional formulas: velocity, acceleration, force, work, pressure.
4. Scientific notation + order of magnitude of: 0.000456; 7,850,000; 299,792,458; 0.0000000082.
5. Rod measured 25.60 cm, true 25.30 cm: (a) absolute (b) relative (c) percentage error.
6. Disc radius (10.0 ± 0.2) cm: % error in area $A = \pi r^2$.
7. A = (50.0 ± 0.5), B = (20.0 ± 0.2), Z = A/B: max absolute and relative error.
8. Dimensional check of $s = ut + \frac{1}{2}at^2$.
9. (story) SI vs Imperial units caused a mission failure — explain and give lessons.
10. Cube side (5.00 ± 0.02) cm: (a) % error in volume (b) max absolute error.
11. (story) Mars Climate Orbiter: what happened, which units were mixed, lessons.

ORIGINAL QUESTIONS — Worksheet: Calculus & 1D (photos m_025, m_028)

1. A particle moves along a straight line with position function $x(t) = 6t^3 - 3t^2 + 6t + 2$ (x in metres, t in seconds). Find (a) velocity as a function of time (b) acceleration (c) velocity and acceleration at t = 2 s.
2. Evaluate the indefinite integral $\int(6x^4 - 8x^3 + 5x^2 - 4x + 7)\,dx$.

3. Velocity of a particle $v(t) = 6t^2 - 4t + 3$ m/s; if it starts from position $x = 2$ m at $t = 0$, determine its position at $t = 3$ s.

4. Differentiate $y = 3x^4 - 5x^2 + 7 \sin x + 4e^x - 6 \ln x$; hence find dy/dx at $x = 1$.

11. (robotics story) End-effector position $x(t) = 2t^3 - 9t^2 + 12t$. (a) derive $v(t)$ and $a(t)$ (b) times at which the end-effector is stationary (c) explain how velocity/acceleration info is useful in robotic motion planning with a real-world example.

12. (variable force) $F(x) = kx$. (a) derive work done moving $x = 0$ to X by integration (b) explain why integration is needed when force is not constant (c) one engineering application; numerical case $k = 500$ N/m, $X = 0.20$ m: calculate the work.

ORIGINAL QUESTIONS — Worksheet: Friction & NLM (g = 9.8 unless stated)

1. 5 kg, $\mu s = 0.40$: maximum force without moving.

2. 10 kg pulled 35 N, $\mu k = 0.20$: (a) normal force (b) friction (c) net force (d) acceleration.

3. 6 kg, $\mu s = 0.50$, $\mu k = 0.30$, $F = 20$ N: does it move? actual friction?

4. 8 kg, $\mu s = 0.40$, $\mu k = 0.25$, force ramped from 0: (a) force at which it starts moving (b) friction just after (c) acceleration.

5. 5 kg, $F = 40$ N at 30° above horizontal, $\mu k = 0.20$: acceleration.

6. 10 kg, $F = 50$ N at 30° below horizontal (push), $\mu k = 0.25$: acceleration.

7. 10 kg on 30° incline, $\mu k = 0.20$, slides down: acceleration.

8. 5 kg on 37° incline, $\mu s = 0.50$ ($g = 10$, $\sin 37 = 0.6$, $\cos 37 = 0.8$): stays or slides?

9. 4 kg on 30° incline, $F = 30$ N up-parallel, $\mu k = 0.20$: acceleration.

10. 6 kg pulled up 30° incline by $T = 50$ N, $\mu k = 0.15$: acceleration.

11. 5 kg on rough table ($\mu k = 0.20$) + 3 kg hanging, frictionless pulley: acceleration and tension.

ORIGINAL QUESTIONS — Worksheet: Circular Motion

1. $r = 2$ m, arc 6 m: angle. 2. 120° in radians. 3. $\theta = 3\pi/4$, $r = 4$: arc length. 4. $\theta = 60°$, $r = 5$: (x, y) coordinates. 5. $r = 3$, $v = 6$: centripetal acceleration. 6. $r = 2$, $\omega = 5$: (a) v (b) a_c. 7. $a_t = 4$, $r = 2$: α. 8. $r = 4$, $\omega = 3$, $\alpha = 2$: (a) a_t (b) a_r (c) total a. 9. 20 rev in 10 s: (a) θ (b) ω. 10. $T = 0.5$ s: ω. 11. $f = 5$ Hz: (a) ω (b) T. 12. 8π rad in 4 s: average ω and revolutions. 13. From rest, $\alpha = 3$, $t = 5$: (a) ω (b) θ. 14. $\omega_0 = 4$, $\alpha = 3$, $t = 6$: ω, θ. 15. $\omega_0 = 20$, $\alpha = -4$ to rest: (a) t (b) θ. 16. $\theta = 20$, ω: 4→12: α. 17. $m = 2$, $r = 4$, $v = 8$: (a) a_c (b) F_c. 18. 1000 kg, $r = 50$, $v = 10$: centripetal force. 19. 5 kg, $\omega = 4$, $r = 2$: (a) v (b) a_c (c) F_c. 20. (explain) centripetal vs centrifugal force. 21. 4 kg on 2 m string at 6 m/s: tension. 22. 2 kg, 1.5 m, 4 m/s: (a) a_c (b) T. 23. 1200 kg, $r = 40$, $v = 12$: centripetal force. 24. 5 kg, $r = 2$ m, $\mu = 0.4$: max speed without slipping. 25. $r = 5$, $\omega = 2$, $\alpha = 3$: (a) a_t (b) a_r (c) total. 26. 3 kg, $r = 2$, 4→8 m/s in 2 s: (a) a_t (b) a_c (c) total. 27. (comment) $r = 100$, $v = 20$: a_c; what if speed doubles? 28. $r = 0.5$, 120 rpm: (a) ω (b) v (c) a_c. 29. $r = 3$, $\theta(t) = 2t^2 + 3t$: (a) ω at $t = 2$ (b) α (c) v (d) a_t (e) a_c. 30. $r = 2$, $\omega(t) = 3t^2 + 2$: (a) α at $t = 2$ (b) v (c) a_t (d) a_c.

ORIGINAL QUESTIONS — Worksheet: Work & Energy (g = 9.8)

1. 5 kg at 4 m/s: KE. 2. $p = 10$, $m = 2$: KE. 3. 5 kg at 10 m: PE. 4. $F = 80$ N over 8 m along force: work. 5. $F = 100$ N over 5 m at 60°: work. 6. Spring $k = 200$, stretch 0.1 → 0.3 m: work by spring. 7. (explain) same work on two paths → what does it mean for the force? 8. 4 kg at rest, net work 200 J: final speed. 9. 5 kg at 4 m/s, 90 J work: final speed. 10. 2 kg free-falls 20 m from rest: speed at ground. 11. 5 kg falls 8 m, loses 50 J to friction: final speed. 12. 10 kg falls 5 m, loses 100 J to friction: final speed. 13. (5-part) 4 kg pushed 30 N through 10 m against 5 N friction from rest: (a) work by push (b) work by friction (c) net work (d) final KE (e) final speed. 14. KE constant, $m \to 4m$: what happens to v? 15. (explain) fall with friction: is mechanical energy conserved?

ORIGINAL QUESTIONS — Worksheet: Collisions & Momentum

1. 2 kg at 6 m/s head-on into stationary 4 kg: total momentum before. 2. 5 kg at 4 + stationary 3, stick: common velocity. 3. 2 kg at 8 + stationary 6, perfectly inelastic: (a) p_i (b) v (c) KE_i (d) KE_f (e) energy lost. 4. 2 kg at 10 elastic into stationary 3: both final velocities. 5. 4 kg at 6 vs 2 kg at 3 opposite: both final velocities (elastic). 6. 5 kg at 8 + stationary 3, stick: (a) v (b) KE_i (c) KE_f (d) lost. 7. 2 kg at 10, stationary 3, e = 0.6: both v. 8. 4 kg at 8, stationary 6, e = 0.5: both v. 9. 3 kg at 6, stationary 2, e = 0.4: approach speed, separation speed, both v. 10. 2 kg at 12, stationary 4, e = 0.5: both v. 11. 4 kg east at 5 + 6 kg north at 3, stick: final speed and direction. 12. 5 kg at +x 8 + 3 kg at +y 6, stick: final speed and direction. 13. Identical balls elastic, first at 8, second rest: final velocities. 14. System mass 10 kg, $a_{cm} = 2$ m/s²: net external force. 15. 2 kg at 10 collides stationary 3 then combined 5 hits stationary 5 (all stick): final velocity. 16. 10 kg at rest explodes into 4 kg at 12 m/s + 6 kg: velocity of 6 kg piece. 17. $F_{net} = 0$ on a system: (a) total momentum (b) velocity of centre of mass (c) acceleration of centre of mass.

Momentum worksheet extras: 5. Uniform rod 6 m, 12 kg from x = 2 to x = 8: CoM. 6. 5 kg at 4 m/s: momentum. 7. 2 kg right at 8 + 4 kg left at 3: total momentum. 8. 0.5 kg at 10 rebounds 6 from wall: (a) p_i (b) p_f (c) Δp (d) impulse. 9. 4 kg at 6 + stationary 2 stick: v. 10. 2 kg east at 6, 3 kg north at 4: (a) p_x (b) p_y (c) magnitude (d) direction. 11. 2 kg east at 5 + 1 kg north at 4 stick: (a) final components (b) speed (c) direction. 12. 2 kg at 4 + 3 kg at 6 same direction: (a) total momentum (b) centre-of-mass velocity. 13. System mass 20, external force 60 N: acceleration of CoM. 14. Mass 10, $a_{cm} = 2$: net force. 15. (double collision, above). 16. (explosion, above). 17. ($F_{net} = 0$, above).

ORIGINAL QUESTIONS — Worksheet: Centre of Mass

1. 2 kg at x = 0, 8 kg at x = 5: x_{cm}. 2. 2 kg at (0,0), 3 kg at (4,0), 5 kg at (10,0): x_{cm}. 3. 2 kg and 8 kg separated by 10 m: CoM measured from the 2 kg mass. 4. 2 kg at (0,0), 3 kg at (6,0), 5 kg at (0,4): CoM point.

Contest-2 (portal): ice cube (m, L) melts in tray (M) — does CoM move, direction, effect of mass ratio (no numbers).

ORIGINAL QUESTIONS — CONTEST-1 printed paper (11 Sep, offline)

Part A (1M MCQ): Q1–2 (page cover, definitions). Q3 Which error from unpredictable variations (options: gross/systematic/random/percentage). Q4 Body from rest, a = 2 m/s², velocity after 5 s (5/10/15/20). Q5 Which is a vector quantity (mass/time/speed/velocity). Q6 Direction of cross product determined by (triangle law/parallelogram law/right-hand rule/head-to-tail). Q7 Rate of change of velocity with time (displacement/speed/acceleration/jerk). Q8 Maximum range of projectile (30/45/60/90). Q9 Significant figures in 0.00456. Q10 SI unit of acceleration. Q11 Vectors of magnitude 5 and 4 perpendicular: dot product. Q12 u = 10, v = 30 in 5 s: average acceleration. Q13 Vector multiplied by negative scalar: what changes.

Part B (2M, answer all): Q16 Order of magnitude: definition + 0.00000487 and 564000000. Q17 Dot vs cross product, at least two differences each. Q18 (1) dy/dx of $y = 4x^2 - 3x + 7$ (2) $\int(3x^2 + 2x)\,dx$. Q19 Two position-time graphs (straight line; upward curve): state how average and instantaneous velocity change in each. Q20 Car A 20 m/s, Car B 12 m/s same direction: velocity of A rel B; then if opposite directions.

Part C (5M): Q21 F = 80.0 ± 0.8 N, a = 25.0 ± 0.5 m/s²: (1) mass (2) max error (3) relative error (4) percentage error. Q22–Q23 (pages not photographed in this set — unknown). Q24 Ball kicked 25 m/s at 37° (g = 10, sin37 = 0.6, cos37 = 0.8): (1) components (2) time to return (3) max height. Q25 Drone: $x(t) = 3t^2 + 4t + 2$, $y(t) = 2t^3 - 3t^2 + 5t$: (1) velocity components and speed at t = 2 (2) acceleration components and magnitude at t = 2 (3) direction of motion vs +x axis. Q26–Q27 (not in photographed set). Unnumbered 5M: $a(t) = 10t + 6$ with $v(0) = 4$, $x(0) = 3$: (1) v(t) (2) x(t) (3) x(1). Unnumbered 5M traffic story: 0→20 m/s in 5 s then constant 20 for 5 s: v-t graph, displacement, average velocity, average acceleration first 5 s.

Part D (10M, answer ONE): Q28a Ball-in-truck: person in back of truck throws ball vertically up at 5.0 m/s relative to truck; truck moves forward at 10.0 m/s with constant acceleration 2.0 m/s²; g = 10; hand fixed relative to truck; catch at same height. (a) Evaluate claim: "ball returns because it already has the truck's horizontal velocity" — correct? (b) Support: initial velocity vector rel ground, time to return, horizontal separation from hand, ahead or behind? (c) Design: horizontal component of launch rel truck (keep upward 5.0) needed to return to hand; does

design change if truck's initial speed doubles with same acceleration? (d) Check: significant digits in the time from (b). Q28b Airplane in wind: 100 km/h through air pointing due north, wind 60 km/h east, east = +î north = +ĵ. (a) Evaluate claim: "point due north → travels due north over ground" — correct? (b) Support: vectors for air velocity and wind, sum = ground velocity, magnitude and direction, displacement after 30 minutes, sketch triangle. (c) Design: required east–west and north–south components of air velocity to travel due north (speed still 100), ground speed, sketch direction (no angle needed). (d) Assess limitation: wind 120 km/h with airspeed 100 — can any heading make it go due north? Justify with components.

WHAT IS NOT IN THIS FILE

- Photos not yet transcribed to text: m_029, m_042–044 (1D motion remainder), m_002/m_049 (projectile), m_050 (relative), m_051/m_054/m_057–059 (NLM), Contest-1 Q22/23/26/27 (pages never photographed or unread), and the freehand notes pages (listed in inventory above).
- The 146 photos themselves live in ap_media/m_001 … m_146 (not public; ask for specific pages).
- Slack live session expired (QR re-login needed); all AP announcements already captured above.